

NCERT Exemplar Solutions for Class 12

Physics

Chapter 8-Electromagnetic Waves

Multiple Choice Questions-I

1. One requires 11eV of energy to dissociate a carbon monoxide molecule into carbon and oxygen atoms. The minimum frequency of the appropriate electromagnetic radiation to achieve the dissociation lies in

- (a) visible region.
- (b) infrared region.
- (c) ultraviolet region.
- (d) microwave region.

Ans: option (c)

$$E = 11 \text{ eV} = 11 \times 1.6 \times 10^{-19} \text{ J}$$

$$h = 6.62 \times 10^{-34} \text{ J-s}$$

$$E = hf$$

By putting the values in the formula:

$$f = \frac{E}{h} = \frac{11 \times 1.6 \times 10^{-19}}{6.62 \times 10^{-34}} = \frac{8.8 \times 10^{-19+34}}{3.31}$$

$$\frac{880}{331} \times 10^{15} \text{ Hz} = 2.65 \times 10^{15} \text{ Hz}$$

Therefore we can see that the above value lies in the range of UV region

2. A linearly polarized electromagnetic wave given as $E_i \cos(kz - \omega t)$ is incident normally on a perfectly reflecting infinite wall at $z = a$. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

- (a) $E_r = -E_0 \hat{i} \cos(kz - \omega t)$.

(b) $E_r = E_0 \hat{i} \cos (kz + \omega t)$.

(c) $E_r = -E_0 \hat{i} \cos (kz + \omega t)$.

(d) $E_r = E_0 \hat{i} \cos (kz - \omega t)$.

Ans: Option (b)

As E is along the +ve X axis so reflected rays will be along the negative X-axis. Its electric component will also be opposite to earlier, i.e. in -z direction, phase will change by $E_r = E_0 (-i) \cos[k(-z) - \omega t + \pi]$

Since $\cos \theta = (\pi - \theta) = -\cos \theta$

$$E_r = -iE_0 \cos[\pi - (kz + \omega t)]$$

$$E_r = iE_0 \cos(kz + \omega t)$$

3. Light with an energy flux of $20W / cm^2$ falls on a non-reflecting surface at normal incidence. If the surface has an area of $30cm^2$, the total momentum delivered (for complete absorption) during 30 minutes is

(a) $36 \times 10^{-5} kg m / s$.

(b) $36 \times 10^{-4} kg m / s$.

(c) $108 \times 10^4 kg m / s$.

(d) $1.08 \times 10^7 kg m / s$.

Ans: Option (b)

Energy flux $= \phi = 20W / cm^2$

Area $A = 30cm^2$, time $= t = 30 \times 60$ sec

$U =$ total energy falling t sec $=$ Energy flux $\times$ Area $\times$ time $= \phi$

$$U = 20 \times 30 \times 30 \times 60 J$$

$$\text{Momentum of the incident light} = \frac{U}{C} = \frac{20 \times 30 \times 30 \times 60}{3 \times 10^8}$$

$$= 36 \times 10^{-4} kg ms^{-1}$$

As there is no reflection from the surface and for complete absorption, the momentum of reflected radiation is zero.

Momentum delivered to surface = Change in momentum

$$p_f = p_i = 0 - 36 \times 10^{-4} = -36 \times 10^{-4} \text{ kg m / s}$$

(-) sign shows the direction of momentum.

4. The electric field intensity produced by the radiations coming from 100W bulb at a 3m distance is E . The electric field intensity produced by the radiations coming from 50W bulb at the same distance is

(a) $\frac{E}{2}$

(b) $2E$

(c) $\frac{E}{\sqrt{2}}$

(d) $\sqrt{2}E$

Ans: Option (c)

$$E_0 \propto \sqrt{P}$$

$$\frac{(E_0)_1}{(E_0)_2} = \sqrt{\frac{(P)_1}{(P)_2}} = \sqrt{\frac{100W}{50W}} = \frac{\sqrt{2}}{1}$$

$$\text{Therefore } (E_0)_2 = \frac{(E_0)_1}{\sqrt{2}}$$

5. If E and B represent electric and magnetic field vectors of the electromagnetic wave, the direction of propagation of electromagnetic wave is along

(a) $\vec{E}$

(b) $\vec{B}$

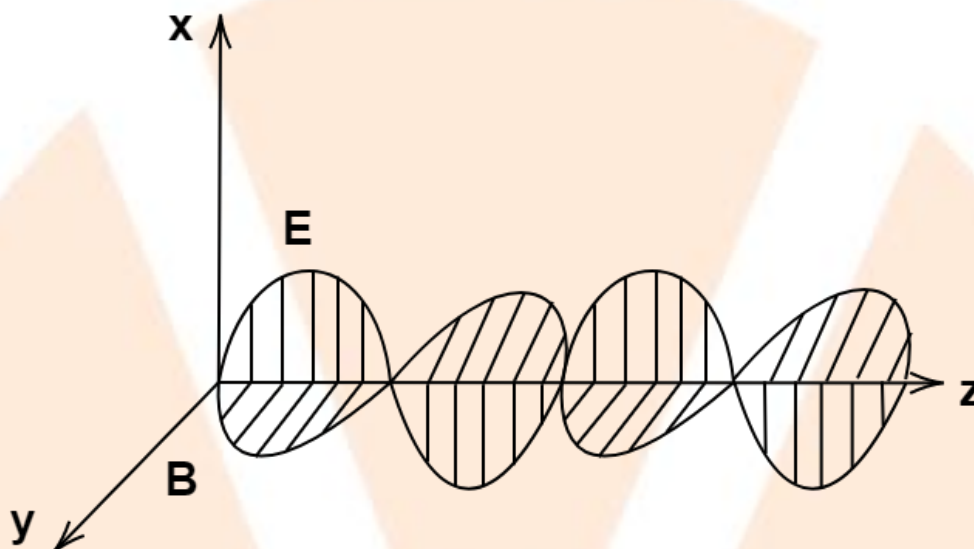
(c) $\vec{B} \times \vec{E}$

(d) $\vec{E} \times \vec{B}$

Ans: Option (d)

The direction of propagation of electromagnetic waves is perpendicular to both $\vec{E}$ and $\vec{B}$ by right thumb rule.

$\vec{E} \times \vec{B}$ by the right thumb rule.



The electric field E is along the $E(+x)$ and $E(-x)$ axis and magnetic field B is along the $B(y)$ and $B(-y)$ axis. So by cross product of E and B, direction is perpendicular to E and B, from and i.e. $(\vec{E} \times \vec{B})$ in $+z$.

6. The ratio of contributions made by the electric field and magnetic field components to the intensity of an EM wave is

- (a) $c : 1$
- (b) $c^2 : 1$
- (c) $1 : 1$
- (d) $\sqrt{c} : 1$

Ans: Option (c)

Average energy by electric field E_0 is U_{av} .

$$U_{av} = \frac{1}{2} \epsilon_0 E_0^2$$

But $E_0 = cB_0$

$$U_{av}, \text{ electric field} = \frac{1}{2} \epsilon_0 (cB_0)^2 = \frac{1}{2} \epsilon_0 c^2 B_0^2$$

$$= \frac{1}{2} \epsilon_0 \frac{1}{\mu_0 \epsilon_0} (B_0)^2, \text{ here } c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$(U_{av})_{\text{electric field}} = \frac{1}{2\mu_0} B_0^2 (U_{av})_{\text{magnetic field}}$$

$$\text{Ratio} = \frac{U_{(av)\text{magnetic field}}}{U_{(av)\text{magnetic field}}} = \frac{1}{1}, \text{ i.e. } 1:1.$$

7. An EM wave radiates outwards from a dipole antenna, with E_0 as the amplitude of its electric field vector. The electric field E_0 which transports significant energy from the source falls off as

(a) $\frac{1}{r^3}$

(b) $\frac{1}{r^2}$

(c) $\frac{1}{r}$

(d) remains constant.

Ans: Option (c)

As we know that electromagnetic waves are radiated from dipole antennas and radiated energy $E \propto \frac{1}{r}$.

Multiple Choice Questions-II (More than one correct)

8. An electromagnetic wave travels in vacuum along +z direction
 $E = (E_1 i + E_0 j) \cos(kz - \omega t)$

Choose the correct options from the following:

(a) The associated magnetic field is given as $B = \frac{1}{C}(E_1 i - E_2 j) \cos(kz - \omega t)$

(b) The associated magnetic field is given as $B = \frac{1}{C}(E_1 i + E_2 j) \cos(kz - \omega t)$

(c) The given electromagnetic field is circularly polarised.

(d) The given electromagnetic wave is plane polarised.

Ans: Options (a) and (d)

We can see that in the given question we have electromagnetic wave, the electric field vector which is given as $E = (E_1 i - E_2 j) \cos(kz - \omega t)$. In electromagnetic wave the associated magnetic field Vector $\vec{B} = \frac{\vec{E}}{C} = \frac{1}{C}(E_1 i + E_2 j) \cos(kz - \omega t)$ verify the option (a) and (b) as $\vec{E}$ and $\vec{B}$ are along X- and Y-axis respectively so we can say that the direction of propagation is Z-axis and the electromagnetic wave is plane polarized along the X and Y direction $\vec{E}$ and $\vec{B}$ respectively verifies option (d).

9. An electromagnetic wave traveling along z-axis is given as $E = E_0 \cos(kz - \omega t)$. Choose the correct options from the following:

(a) The associated magnetic field is given as $B = \frac{1}{c} k \times E = \frac{1}{\omega}(k \times E)$

(b) The electromagnetic field can be written in terms of the associated magnetic field as $\vec{E} = c(\vec{B} \times \vec{K})$

(c) $\vec{K} \times E = 0 \times B \times \vec{K} = 0$

(d) $\vec{K} \times E = 0 \times \vec{B} \times \vec{K} = 0$

Ans: Options (a), (b) and (c)

In the above question we can see that the electromagnetic wave is traveling in +Z direction. Therefore we can say that its electric field is given by $E = E_0 \cos(kz - \omega t)$ along X direction which is perpendicular to Z-axis i.e., along X direction. Thus it proves that the associated magnetic field B is also perpendicular to +Y.

10. A plane electromagnetic wave propagating along x direction can have the following pairs of E and B

- (a) E_x, B_y
- (b) E_y, B_z
- (c) B_x, E_y
- (d) E_z, B_y

Ans: Options (b) and (d)

We can see that in the above question the EM wave is plane polarized and thus its propagation is in +X direction. So we can say that it will be in either Y and Z direction or Z and Y direction. Thus it verifies the direction of $\vec{E}$ and $\vec{B}$ will be in either Y and Z direction or Z and Y direction. So this verifies answers (b) and (d).

11. A charged particle oscillates about its mean equilibrium position with a frequency of 10^9 Hz . The electromagnetic waves produced:

- (a) will have frequency of 10^9 Hz
- (b) will have frequency of $2 \times 10^9 \text{ Hz}$
- (c) will have wavelength of 0.3 m
- (d) fall in the region of radio waves.

Ans: Options (a) (c) and (d)

Vibrating particles produce electric and magnetic fields, so will produce an E.M. wave of same frequency 10^9 Hz verifies answer (a).

$$V = 10^9 \text{ Hz}, c = 3 \times 10^8 \text{ m/s}$$

$$\text{So } \lambda = \frac{c}{v} = \frac{3 \times 10^8}{10 \times 10^8} = 0.3 \text{ m}$$

Therefore it verifies answer (c).

As the range of radio waves is 10 Hz to 10^{12} Hz and 10^9 Hz lies between this ranges. Hence, option (d) is verified.

12. The source of electromagnetic waves can be a charge

(a) moving with a constant velocity.

(b) moving in a circular orbit.

(c) at rest.

(d) falling in an electric field.

Ans: Option (b)

We know that an electromagnetic wave can be produced either by accelerated or by oscillating charge. Here, we can see that in option (b) charge is moving in a circular orbit. In circular motion, the direction of the motion of charge is changing continuously, thus it proves our explanation that it is an accelerated motion and this option is correct.

13. An E.M. wave of intensity I falls on a surface kept in vacuum and exerts radiation pressure p on it. Which of the following are true?

(a) Radiation pressure is $\frac{I}{c}$

if the wave is totally absorbed

(b) Radiation pressure is $\frac{I}{c}$

if the wave is totally reflected

(c) Radiation pressure is $\frac{2I}{c}$ if the wave is totally reflected.

(d) Radiation pressure is in the range $\frac{1}{c} < p < \frac{2I}{c}$ for real surfaces.

Ans: Options (a) (c) and (d)

We know that the radiation pressure is the force exerted by particles (dual nature of particle) on unit area, due to the change in momentum of radiated particles per unit area per sec = $\frac{1}{c}$

Here we have

I = intensity of radiation

C = velocity of radiation

We know that when the wave is fully absorbed by the surface, then the momentum of the reflected wave per unit time per unit area is zero.

Radiations are absorbed, so momentum per unit area per second $= \frac{1}{c}$ thus it verifies the first option i.e. (a).

When radiation is reflected back, the momentum becomes double as in the earlier case, so discards the option (b) and it verifies the third option i.e. (c). So variation of radiation pressure p comes between the range $\frac{1}{c} < p < \frac{2I}{c}$ thus it verifies the option (d).

Very Short Answer Type Questions

14. Why is the orientation of the portable radio with respect to the broadcasting station important?

Ans: Transmitted carrier wave signals are plane polarized and if the intensity of signal is poor then the receiving antenna of the radio must be parallel to the component of either electric or magnetic field. Because energy is only due to amplitudes of electric and magnetic components in EM waves, magnitude of amplitude is in particular direction perpendicular to each other and perpendicular to wave propagation.

15. Why does the microwave oven heat up a food item containing water molecules most efficiently?

Ans: We know that in Microwave oven, molecules of food items start to vibrate by a driven force due to the frequency of the microwave. But when the natural frequency of water molecules matches with microwave frequency then it causes resonance (more amplitude) which further causes increase in temperature.

16. The charge on a parallel plate capacitor varies as $q = q_0 \cos 2\pi vt$. The plates are very large and close together (area = A , separation = d). Neglecting the edge effects, find the displacement current through the capacitor?

Ans: The displacement current I_d in capacitor is

$$I_d = I_c = \frac{dq}{dt}, \text{ Where } q = q_0 \cos(2\pi vt) \text{ (Given)}$$

$$I_d = I_c = \frac{d}{dt} q_0 \cos(2\pi vt)$$

$$I_d = q_0 2\pi v \sin \pi vt$$

$$I_d = 2\pi v q_0 \pi v t$$

17. A variable frequency a.c source is connected to a capacitor. How will the displacement current change with decrease in frequency?

Ans: As we know that $X_c = \frac{1}{2\pi VC}$ and $I = \frac{V}{X_c}$

So reactance of capacitor increases on decreasing frequency $X_c = \frac{1}{v}$

$$\text{So } I = 2\pi v CV$$

As the reactance of the capacitor increases, the current by Ohm's law will decrease.

$$I \propto \frac{1}{X_c}$$

So the displacement current decreases and frequency decreases when the conduction current is equal to displacement current.

18. The magnetic field of a beam emerging from a filter facing a floodlight is given by $B = 12 \times 10^{-8} \sin (1.20 \times 10^7 z - 3.6 \times 10^{15} t) T$. What is the average intensity of the beam?

Ans: Let the first equation be $B = 12 \times 10^{-8} \sin(1.20 \times 10^7 z - 3.6 \times 10^{15} t)T$. Standard equation $B = B_0(kz - \omega t)$

Comparing the different terms in above two (I, II) equations

$$B_0 = 12 \times 10^{-8} \text{ Tesla}$$

$$I_{av} = \frac{B_0^2}{2\mu_0} = \frac{12 \times 10^{-8} \times 12 \times 10^{-8} \times 3 \times 10^8}{2 \times 4\pi \times 10^{-7}}$$

$$\text{It gives} = \frac{54}{3.14 \times 10} = 1.72 \text{ W / m}^2$$

19. Pointing vectors S is defined as a vector whose magnitude is equal to the wave intensity and whose direction is along the direction of wave propagation. Mathematically, it is given by $S = \frac{1}{\mu_0}(\bar{E} \times \bar{B})$. Show the nature

of S versus t graph.

Ans: Consider an electromagnetic wave. Let electric field (E) of EM wave varies along Y-axis the propagation of wave is along X-axis, then $\bar{E} \times \bar{B}$ will give the direction of flow of energy in electromagnetic wave along X= axis

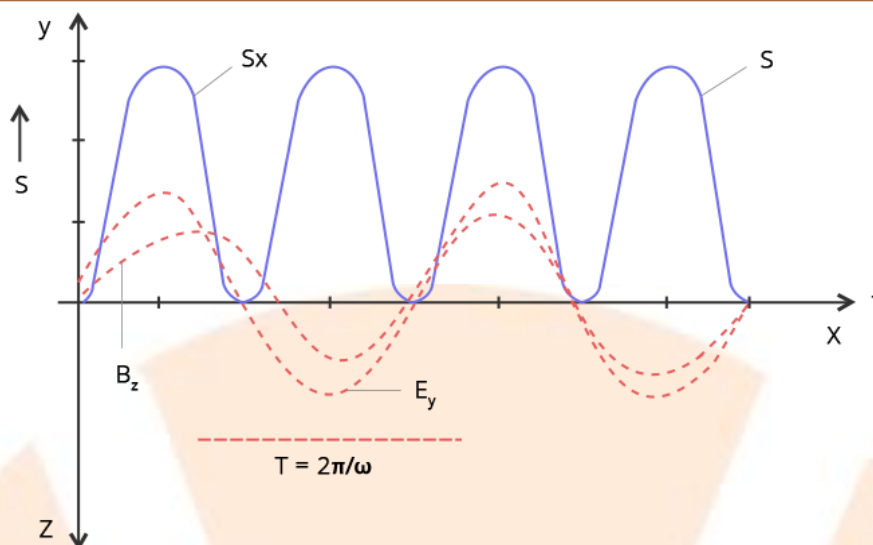
$$\bar{E} = E_0 \sin(\omega t - kx) \hat{j}$$

$$\bar{B} = B_0 \sin(\omega t - kx) \hat{k}$$

$$S = \frac{1}{\mu}(\bar{E} \times \bar{B})$$

$$S = \frac{1}{\mu_0} \sin^2(\omega t - kx) \hat{i}$$

Variation of |S| with time will be as given in figure.



20. Professor C.V Raman surprised his students by suspending a tiny light ball in a transparent vacuum chamber by shining a laser beam on it. Which property of EM waves was he exhibiting? Give one more example of this property

Ans: In this question it is given that Professor C.V Raman once surprised his students by suspending a tiny light ball freely in a transparent vacuum chamber by shining a laser beam on it. We know that radiation and matter possess dual nature. Also we know the property which states that an EM wave carries energy and momentum. Due to this change in momentum (by direction or velocity of wave), EM wave exerts pressure on the surface, by reflection and refraction. There this was the property of EM waves that helped Professor CV Raman to surprise his students by suspending a tiny light ball freely in a transparent vacuum chamber by shining a laser beam on it. We should note that the tails of the comets are also due to radiation pressure. Electromagnetic radiation can pass even through vacuum and has particle nature.

Therefore, mobile phones placed in an evacuated transparent chamber can ring up which can be seen through the transparent chamber, but sound of ring tone cannot be heard, hence it proves the propagation of electromagnetic waves in vacuum.

21. Show that the magnetic field B at a point in between the plates of a parallel-plate capacitor during charging is $\frac{\mu_0 \epsilon_0 r}{2} \frac{dE}{dt}$ (symbols having usual meaning).

Ans: Let I_d be the displacement current in the region of magnetic field between two plates of parallel plate capacitor. The magnetic field induction at a point in a region between two plates of a capacitor at a perpendicular distance from the axis of plate is

$$B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0}{2\pi r} I_d$$

$$B = \frac{\mu_0 \epsilon_0}{2\pi r} = \frac{d}{dt} (E \pi r^2)$$

Therefore it gives the value:

$$B = \frac{\mu_0 \epsilon_0 r}{2} \frac{dE}{dt}$$

Short Answer Type Questions

22. Electromagnetic waves with wavelength

- (i) λ_1 is used in satellite communication.
 - (ii) λ_2 is used to kill germs in water purifiers.
 - (iii) λ_3 is used to detect leakage of oil in underground pipelines.
 - (iv) λ_4 is used to improve visibility in runways during fog and mist conditions.
- (a) Identify and name the part of the electromagnetic spectrum to which these radiations belong.

Ans: (i) Microwave is used in satellite communications so, λ_1 is the wavelength of the microwave. It is used in a microwave oven.

(ii) Ultraviolet rays are used to kill germs in water purifiers so, λ_2 is the wavelength of ultraviolet rays.

(iii) X-rays are used to detect leakage of oil in underground pipelines so λ_3 is in the X-ray region. It is also used to detect cracks in machinery and to detect fractures in bones of the body.

(iv) Infrared rays λ_4 are used to improve visibility due to larger wavelengths of low scattering. Infrared rays are used in optical communication.

(b) Arrange these wavelengths in ascending order of their magnitude.

Ans: The arrangement of the wavelengths in ascending order are $\lambda_3, \lambda_4, \lambda_2, \lambda_1$

(c) Write one more application for each.

Ans: Uses are along with part (a).

23. Show that the average value of radiant flux density 'S' over a single period 'T' is given by $S = \frac{1}{2c\mu_0} E_0^2$.

Ans: Radiant flux density $S = \frac{1}{\mu_0} (\bar{E} \times \bar{B})$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\text{Or, } c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\Rightarrow \frac{1}{\mu_0} = \epsilon_0 c^2$$

So, the above expression can be written as $S = \epsilon_0 c^2 (\bar{E} \times \bar{B}) \dots (i)$

Let electromagnetic waves be propagated along the X-axis so its electric and magnetic field vectors are along the Y and Z axis.

$$\bar{E} = E_0 \cos(kx - \omega t) \hat{j}$$

$$\Rightarrow \bar{B} = B_0 \cos(kx - \omega t) \hat{k}$$

Now by multiplying both the equation:

$$(\bar{E} \times \bar{B}) = (E_0 B_0) \cos^2(kx - \omega t) (\hat{j} \times \hat{k})$$

Put $E \times B$ in (i)

$$\Rightarrow S_{av} = c^2 \epsilon_0 E_0 B_0 \left[\frac{1}{T} \int_0^T \cos^2(kx - \omega t) dt \right] \hat{i}$$

$$\Rightarrow \frac{c^2 \epsilon_0 E_0 B_0}{T} \left[\frac{T}{2} \right] = \frac{c^2}{2} \epsilon_0 E_0 \left(\frac{E_0}{c} \right)$$

$$\Rightarrow \frac{c \epsilon_0 E_0}{2}$$

$$\Rightarrow \frac{c}{2} \times \frac{1}{\mu_0 c^2} E_0^2$$

Hence it is proved that $S_{av} = \frac{E_0^2}{2\mu_0 c}$

24. You are given a $2\mu F$ parallel plate capacitor. How would you establish an instantaneous displacement current of 1mA in the space between its plates?

Ans: $C = 2F = 2 \times 10^{-6} F$, $I_d = 1 mA = 10^{-3} A$

Therefore, $q = CV$

$$I_d dt = C.dV$$

$$\Rightarrow I_d = C \frac{dV}{dt}$$

$$\Rightarrow \frac{dV}{dt} = \frac{I_d}{C} = \frac{10^{-3} A}{2 \times 10^{-6}}$$

$$= \frac{1000}{2} \text{Volts}$$

$$\therefore \frac{dV}{dt} = 500V / s$$

25. Show that the radiation pressure exerted by an EM wave of intensity I on a surface kept in vacuum is $\frac{1}{C}$.

Ans: Pressure on surface due to particle nature of wave $P = \frac{F}{A}$

Force = Rate of change of momentum

If mass (m) of radiant particle for wave of velocity c then

$$\Rightarrow E = mc^2$$

$E = U$ (let)

$$\Rightarrow U = mc \cdot c$$

$$U = p \cdot c$$

Where, momentum $p = mc$

Now differentiating w.r.t. time (t) we get

$$\Rightarrow \frac{dU}{dt} = c \times \frac{dp}{dt}$$

By Newton's second law,

$$\frac{1}{c} \frac{dU}{dt} = \frac{dp}{dt}$$

Therefore, $F = \frac{1}{c} \frac{dU}{dt}$ or, $\frac{F}{A} = \frac{1}{A} \times \frac{1}{c} \frac{dU}{dt}$

Pressure (P) on the surface due to e.m. wave radiation and $P = \frac{F}{A}$

$$P = \frac{1}{c} \left[\frac{dU}{A dt} \right]$$

We know that intensity of radiation is equal to the radiant energy (U) falling on the unit surface per second.

$$I = \frac{1}{A} \frac{dU}{dt} \text{ or, } P = \frac{1}{c}$$

Hence it is proved.

26. What happens to the intensity of light from a bulb if the distance from the bulb is doubled? As a laser beam travels across the length of a room, its intensity essentially remains constant. What geometrical characteristic of LASER beam is responsible for the constant intensity which is missing in the case of light from the bulb?

Ans: In this question we have to tell what will happen to the intensity of light from a bulb if we double the distance from the bulb. We know that the bulb spreads its light in all ground spherically and symmetrically. So if the distance from the bulb is doubled, then the surface area covered by radiations change from $4\pi r^2$ to $4\pi(2r)^2$ i.e., 2^2 or four times decreased in straight laser.

But we know that in a bulb it decreases by $4(4\pi)$, i.e 16π decreases. So the intensity becomes one fourth the initial value in a straight line. Since $I \propto \frac{1}{r^2}$ and for spherical value $I \propto \frac{1}{4\pi r^2}$.

In the case of a laser, it does not spread in all directions. It passes only along a straight line. So its intensity remains almost the same. Also we know that geometrical characteristics of LASER beam which is responsible for the constant intensity

are:

- (i) Monochromatic,
- (ii) Coherent,
- (iii) Highly collimated from all around source,
- (iv) Unidirectional.

Therefore these characteristics are missing in the case of light from the bulb.

27. Even though an electric field E exerts a force qE on a charged particle yet the electric field of an EM wave does not contribute to the radiation pressure (but transfer energy). Explain.

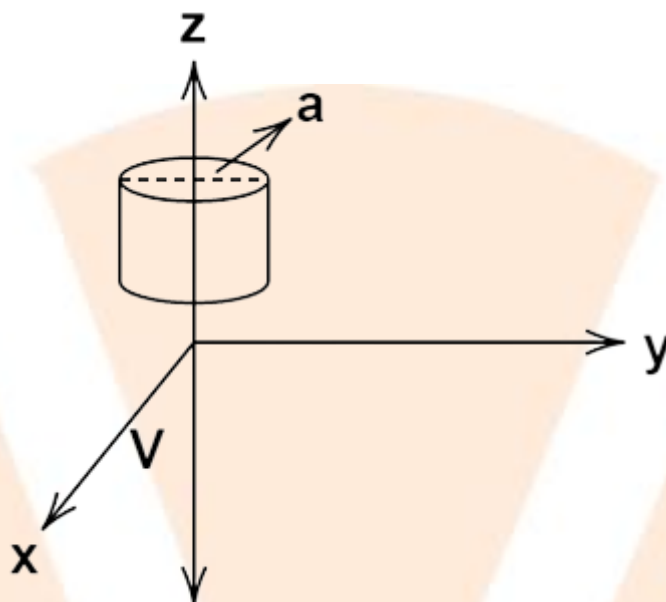
Ans: In electromagnetic waves, the electric field is oscillating, so the resultant electric force on the particle will be zero, as the direction of electric force changes every half of the time period. As the electric field vibrates, the radiation of energy will take place only due to vibrating electric and magnetic fields.

Long Answer Type Questions

28. An infinitely long thin wire carrying a uniform linear static charge density λ is placed along the z-axis . The wire is set into motion along its

length with a uniform velocity $\vec{V} = VK_z$. Calculate the poynting vector

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$



Ans: Consider a cylindrical Gaussian surface in such a way that the axis of the cylinder lies on wire. Electric field intensity due to long straight wire at a distance a and charge density $\lambda C/m$.

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 a} = \frac{\lambda}{2\pi\epsilon_0 a} \hat{j}$$

$$\vec{B} = \frac{\mu_0 I}{2\pi a} \hat{i}$$

$$\Rightarrow I = \frac{q}{t} = \frac{\lambda l}{t} = \lambda v$$

$$\text{Therefore } \vec{B} = \frac{\mu_0 \lambda v}{2\pi a} \hat{i}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{1}{\mu_0} \left[\frac{1}{2\pi\epsilon_0 a} \hat{j} \times \frac{\mu_0 \lambda v}{2\pi a} \hat{i} \right]$$

$$\vec{S} = \frac{-\lambda^2 v}{4\pi^2 a^2 \epsilon_0} \hat{k}$$

29. Sea water at frequency $\nu = 4 \times 10^8 \text{ Hz}$ has permittivity $\epsilon = 80\epsilon_0$, permeability $\mu \approx \mu_0$ and resistivity $\rho = 0.25 \Omega m$. Imagine a parallel plate capacitor immersed in sea water and driven by an alternating voltage source $V(t) = V_0 \sin(2\pi\nu t)$. What fraction of the conduction current density is the displacement current density?

Ans: Suppose distance between the parallel plates of capacitor is 'd' and the applied voltage $V(t) = V_0 \sin(2\pi\nu t)$.

$$E = \frac{V(t)}{d} = \frac{V_0 \sin(2\pi\nu t)}{d}$$

By Ohm's conduction current density

$$\bar{J}_c = \frac{1}{\rho} \bar{E} = \frac{1}{\rho} \frac{V_0 \sin(2\pi\nu t)}{d}$$

$$\text{Let us assume that } J_0^c = \frac{V_0}{\rho d}$$

The current displacement is given as:

$$J_d = \epsilon \frac{dE}{dt} = \epsilon \frac{d}{dt} \frac{V_0}{d} \sin(2\pi\nu t)$$

$$\Rightarrow J_d = \frac{2\pi\nu\epsilon V_0}{d} \cos(2\pi\nu t)$$

$$\text{Again assume that } J_0^d = \frac{2\pi\nu\epsilon V_0}{d}$$

$$\text{Then we have: } J_d = J_0^d \cos(2\pi\nu t)$$

$$\Rightarrow \frac{J_0^d}{J_0^c} = 2\pi\nu\epsilon_0\rho$$

$$\Rightarrow \frac{J_0^d}{J_0^c} = 2\pi\nu\epsilon\rho = 2\pi \times 4 \times 10^8 \times 80\epsilon_0 \times 0.25$$

$$= 4\pi\epsilon_0 \times 2 \times 80 \times 0.25 \times 10^8$$

$$= \frac{160 \times 25}{9 \times 10 \times 100}$$

Therefore, it gives the answer $\Rightarrow \frac{J_0^d}{J_0^c} = \frac{4}{9}$

30. A long straight cable of length l is placed symmetrically along z -axis and has radius a ($a \ll l$). The cable consists of a thin wire and a coaxial conducting tube. An alternating current $I(t) = I_0 \sin(2\pi vt)$ flows down the central thin wire and returns along the coaxial conducting tube. The induced electric field at a distance s from the wire inside the cable is

$$\vec{E}(s, t) = \mu_0 I_0 v \cos(2\pi vt) \log_e \left(\frac{s}{a} \right) \hat{k}$$

(i) Calculate the displacement current density inside the cable.

Ans: Induced electric field $E(s, t)$ at distance s ($s < \text{radius of co-axial cable}$) is given as $E(s, t) = \epsilon_0 I_0 v \cos 2\pi vt \log_e \left(\frac{s}{a} \right) k$.

Displacement current density J_d is given by

$$\begin{aligned} J_d &= \epsilon_0 \frac{dE}{dt} = \epsilon_0 \mu_0 I_0 v \frac{d}{dt} \left[\cos 2\pi vt \log_e \frac{s}{a} \right] k \\ \Rightarrow J_d &= -\epsilon_0 \mu_0 I_0 v \cos 2\pi vt^2 \left(\log_e \frac{s}{a} \right) \sin(2\pi vt) \hat{k} \\ &= -\frac{1}{c^2} I_0 2\pi v^2 \left[-\log_e \left(\frac{a}{s} \right) \right] \sin(2\pi vt) k \\ &= \frac{2\pi v^2}{v^2 \lambda^2} I_0 \log_e \left(\frac{a}{s} \right) \sin 2\pi vt \hat{k} \end{aligned}$$

Therefore this is the required answer.

(ii) Integrate the displacement current density across the cross-section of the cable to find the total displacement current I_d

$$\text{Ans: } I_d = \int J_d s ds d\theta = \int_{s=0}^a J_d s ds \int_0^{2\pi} d\theta = \int_{s=0}^a J_d s ds (2\pi)$$

$$I_d = \left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin(2\pi vt) \hat{k} \int_{s=0}^a \log\left(\frac{a}{s}\right) s ds$$

Now, we will calculate the integration of $\int_{s=0}^a \left(\log \frac{a}{s}\right) s ds$

$$= \left[\log\left(\frac{a}{s}\right) \int_s^a ds \right]_0^a - \int_{s=0}^a \frac{d}{ds} \left[\log\left(\frac{s}{a}\right) \right] \int_s^a s ds ds$$

$$= \left[\log\left(\frac{a}{a}\right) \frac{a^2}{2} - 0 \right] - \frac{1}{2a} \int_{s=0}^a s^3 ds$$

$$= 0 - \frac{1}{2a} \left[\frac{s^4}{4} \right]_0^a = \frac{-a^3}{8}$$

Therefore, $I_d = -\frac{a^3}{8} \left(\frac{2\pi}{\lambda}\right)^2 \sin 2\pi vt \hat{k}$

$$= -\frac{a}{2} \times \frac{a^4}{4} \left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin(2\pi vt) \hat{k}$$

The negative sign shows that the displacement current I_d is opposite to the conduction current I_c

$$I_d = \frac{a}{2} \left(\frac{\pi a}{\lambda}\right)^2 I_0 \sin(2\pi vt) (-k)$$

I_d is in $-Z$ direction as the IC is in $+Z$ direction.

(iii) Compare the conduction current I_0 with the displacement current I_0^d .

Ans: $I_d = \frac{a}{2} \left(\frac{\pi a}{\lambda}\right)^2 I_0 \sin(2\pi vt) (-k)$

$$I_d = I_0^d \sin(2\pi vt)$$

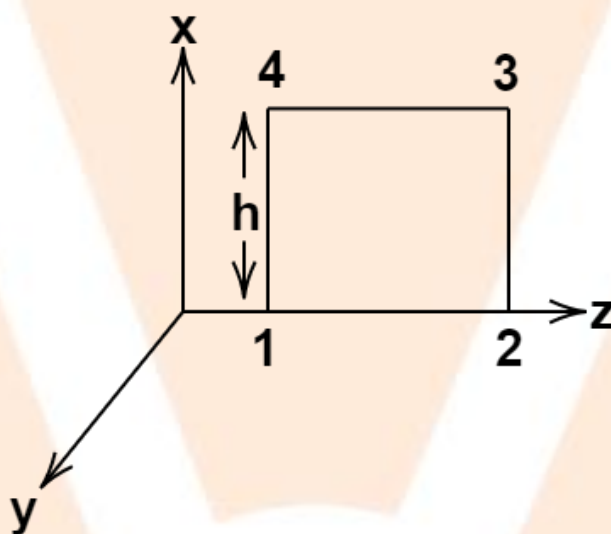
Where it is given that $I_0^d = \frac{a}{2} \left(\frac{\pi a}{\lambda}\right)^2 I_0$

Therefore, the required ratio is $\frac{I_0^d}{I_0} = \frac{\pi^2 a^3}{2\lambda^2}$.

31. A plane EM wave travelling in vacuum along z-direction is given by $\vec{E} = E_0 \sin(kz - \omega t) \hat{i}$ and $\vec{B} = B_0 \sin(kz - \omega t) \hat{j}$

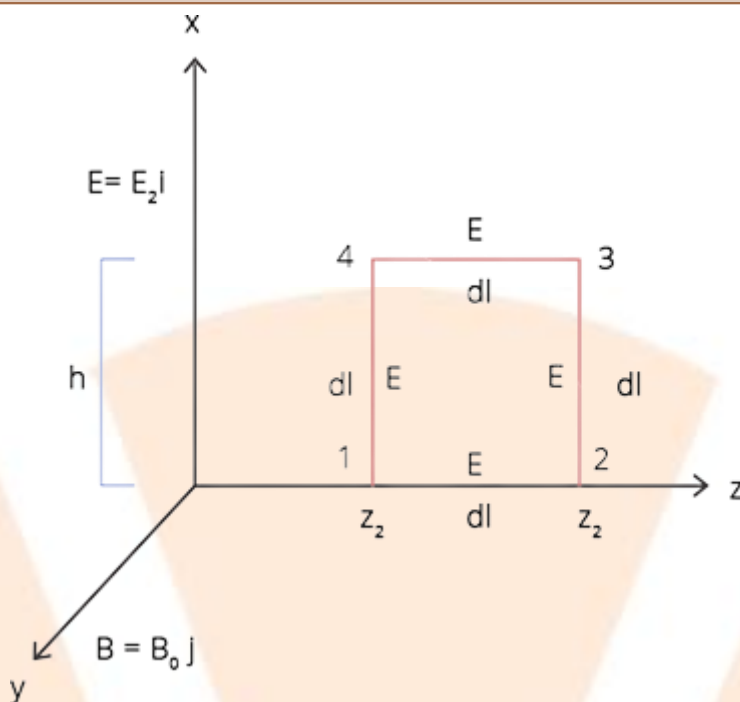
(i) Evaluate $\oint \vec{E} \cdot d\vec{l}$ over the rectangular loop 1234 shown

Ans: As the electromagnetic wave is propagating along the Z-axis then its electric and magnetic field vectors are along the X and Y axis.



(ii) Evaluate $\int \vec{B} \cdot d\vec{s}$ over the surface bounded by loop 1234.

Ans: $\vec{E} = E_0 \hat{i}$ and $\vec{B} = B_0 \hat{j}$ in the figure given below:

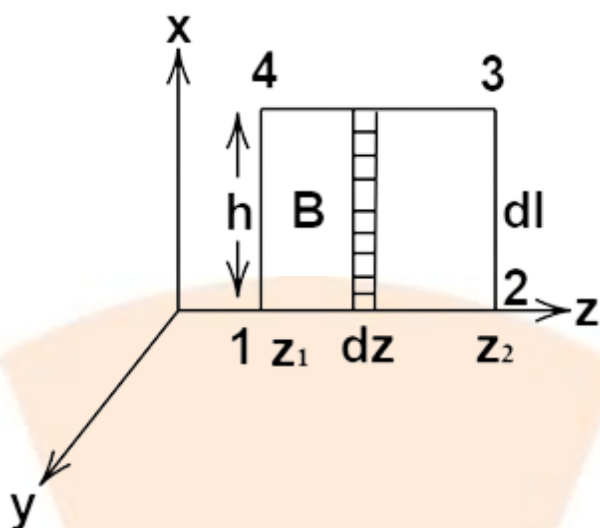


By using the line integral of E over loop 1234,

$$\begin{aligned}\oint \vec{E} \cdot d\vec{l} &= \int_1^2 \vec{E} \cdot d\vec{l} + \int_2^3 \vec{E} \cdot d\vec{l} + \int_3^4 \vec{E} \cdot d\vec{l} + \int_4^1 \vec{E} \cdot d\vec{l} \\ &= \int_1^2 E dl \cos 90^\circ + \int_2^3 E dl \cos 0^\circ + \int_3^4 E dl \cos 90^\circ + \int_4^1 E dl \cos 180^\circ \\ &\Rightarrow 0 + [Eh]_{z_2} - [Eh]_{z_1} = E_0 h \sin(kz_2 - \omega t) - E_0 h \sin(kz_1 - \omega t) \\ &= hE_0 [\sin(kz_2 - \omega t) - \sin(kz_1 - \omega t)]\end{aligned}$$

(iii) Use equation $\int E dl = \frac{-d\phi_B}{dt}$ to prove $\frac{E_0}{B_0} = C$

Ans: Consider a strip of area $ds = h \cdot dz$ as in figure. Angle between $\vec{ds}$ and $\vec{B}$ is zero



$$\int \vec{B} \cdot d\vec{s} = \int \vec{B} \cdot d\vec{s} \cos 0^\circ \hat{j}$$

$$= \int_{z_1}^{z_2} B \cdot ds \hat{j}$$

$$= \int_{z_1}^{z_2} B_0 \sin(kz - \omega t) h \cdot dz$$

$$\int \vec{B} \cdot d\vec{s} = \frac{B_0 h}{k} (\cos(kz_1 - \omega t) - \cos(kz_2 - \omega t))$$

$$\text{Given } \oint E \cdot d\vec{l} = \frac{-d\phi_B}{dt} = -\frac{d}{dt} \oint B \cdot d\vec{s}$$

$$= \frac{d}{dt} \left[\frac{B_0 h}{k} \cos(kz - \omega t) - \cos(kz - \omega t) \right] \hat{j}$$

$$= \frac{B_0 h}{k} [-\sin(kz_1 - \omega t)(-\omega) + \sin(kz_2 - \omega t)(-\omega)]$$

$$E = \frac{B_0 h \omega}{k} [-\sin(kz_1 - \omega t) - \sin(kz_2 - \omega t)]$$

$$E = E_0 [\sin(kz_1 - \omega t) - \sin(kz_2 - \omega t)]$$

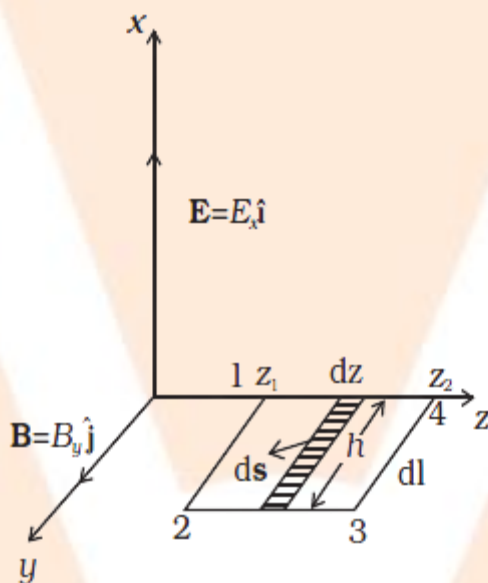
$$\Rightarrow E_0 = B_0 C$$

$$\Rightarrow \frac{E_0}{B_0} = \frac{E_0}{BY} = \frac{B_0 C}{B_0}$$

Therefore it gives the required answer: $\frac{E_0}{B_0} = C$

(iv) By using similar process and the equation $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I + \epsilon_0 \frac{d\phi_E}{dt}$, prove that $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$.

Ans: Consider a loop 1,2,3,4 in Y-Z plane as in figure



$$\begin{aligned}\oint \mathbf{B} \cdot d\mathbf{l} &= \int_1^2 \mathbf{B} \cdot d\mathbf{l} + \int_2^3 \mathbf{B} \cdot d\mathbf{l} + \int_3^4 \mathbf{B} \cdot d\mathbf{l} + \int_4^1 \mathbf{B} \cdot d\mathbf{l} \\ &= \int_1^2 \mathbf{B} \cdot d\mathbf{l} \cos 0^\circ + \int_2^3 \mathbf{B} \cdot d\mathbf{l} \cos 90^\circ + \int_3^4 \mathbf{B} \cdot d\mathbf{l} \cos 180^\circ + \int_4^1 \mathbf{B} \cdot d\mathbf{l} \cos 90^\circ\end{aligned}$$

$$B_0 h [\sin(kz_1 - \omega t) - \sin(kz_2 - \omega t)]$$

Now to evaluate, we will consider the rectangle 1234 to be made of strips of area of $h dz$ each. Therefore we have:

$$\begin{aligned}\Rightarrow \int \mathbf{E} \cdot d\mathbf{s} &= \int \mathbf{E} \cdot d\mathbf{s} \cos 0^\circ = \int \mathbf{E} \cdot d\mathbf{s} = \int_z^{z_2} E_0 \sin(kz - \omega t) h dz \\ &= -E_0 h [\cos(kz_2 - \omega t) - \cos(kz_1 - \omega t)]\end{aligned}$$

Now we will use the relations that we have obtained in the above equation and it gives us

$$\Rightarrow B_0 = E_0 \frac{\omega}{k} \mu \epsilon_0$$

$$\frac{E_0 \omega}{B_0 k} = \frac{1}{\mu_0 \epsilon_0}$$

But we have $\omega = c k$

$$\text{Therefore it gives us the answer } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

32. A plane EM wave travelling along z direction is described by $\vec{E} = E \sin(kz - \omega t) \hat{i}$ and $\vec{B} = B_0 \sin(kz - \omega t) \hat{j}$. Show that

(i) The average energy density of the wave is given by $U_{av} = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4 \mu_0} B_0^2$

Ans: The electromagnetic wave carries energy which is due to the electric field vector and magnetic field vector. In electromagnetic waves, E and B vary with time. E and B varies with time. The energy density due to electric field $\vec{E}$ is

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

The energy density due to magnetic field B is $U_B = \frac{1}{2 \mu_0} B^2$

Total average energy density of electromagnetic wave

$$U_E + U_B = \frac{1}{2 \mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2$$

$$\Rightarrow E = E_0 \sin(kz - \omega t) \hat{i}$$

$$\Rightarrow B = B_0 \sin(kz - \omega t) \hat{j}$$

Average value of E^2 over a cycle $\frac{E_0^2}{2}$

$$\text{Therefore, it gives the answer } U_{av} = \frac{1}{4} \left[\epsilon_0 E_0^2 + \frac{B_0^2}{\mu_0} \right]$$

(ii) The time averaged intensity of the wave is given by $I_{av} = \frac{1}{2} C \epsilon_0 E_0^2$

Ans: We know that $E_0 = CB_0$ and $C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$\frac{1}{4} \frac{B_0^2}{\mu_0} = \frac{1}{4} \frac{E_0^2 / C^2}{\mu_0} = \frac{E_0^2}{4\mu_0} \mu_0 \epsilon_0 = \frac{1}{4} \epsilon_0 E_0^2$$

$$U_B = U_E$$

$$U_{av} = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4} \frac{B_0^2}{\mu_0}$$

Therefore, it gives $\frac{E_0}{B_0} = C$

$$\Rightarrow \frac{E_0^2}{B_0^2} = \frac{1}{\mu_0 \epsilon_0} \Rightarrow B_0^2 = \mu_0 \epsilon_0 E_0^2$$

$$\Rightarrow U_{av} = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4} \epsilon_0 E_0^2$$

$$\Rightarrow U_{av} = \frac{1}{4} \epsilon_0 E_0^2$$

$$\Rightarrow U_{av} = \frac{1}{4} \epsilon_0 \frac{B_0^2}{\epsilon_0 \mu_0} + \frac{1}{4} \frac{B_0^2}{\mu_0}$$

$$= \frac{B_0^2}{4\mu_0} + \frac{B_0^2}{4\mu_0}$$

Therefore $U_E = U_B$

Time average intensity of wave

$$I_{av} = U_{av} \cdot C = \frac{1}{2} \frac{B_0^2 C}{\mu_0} = \frac{1}{2} \epsilon_0 E_0^2 C$$